🍎 Food & Digestion: Reading · ANSWER KEY

Biology A · Unit 4 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Digestion, A4 food molecules notes + reading + Food Cards, digestive system reading + diagram

Name: _____

What Happens to Your Lunch

Food is made of three main macromolecules. Carbohydrates are sugars and starches, built from glucose. Lipids are fats and oils, built from glycerol and fatty acids. Proteins are chains of amino acids; there are 20 kinds. Each big molecule is made of smaller building blocks.

Starch is hundreds of glucose molecules linked together; plants store energy this way. Animals store glucose as glycogen. Cellulose (fiber) is also made of glucose, but our bodies cannot digest it, so it passes through. Extra carbohydrates and extra protein can be stored as fat.

Digestion breaks big food molecules into building blocks small enough to enter the blood. It starts in the mouth: teeth chew and saliva from the salivary glands begins to break down starch. Muscles in the esophagus push food to the stomach, a muscle bag that churns food with acid and enzymes.

Digestion finishes in the small intestine, where nutrients pass into the blood. The liver makes bile to help break up fats, and the pancreas makes digestive enzymes and the hormone insulin. The large intestine absorbs water, and waste leaves the body.

Knowing the building blocks helps explain diabetes later in this unit: carbohydrates become glucose in the blood, and the body has to manage that glucose.

Check your reading

1. What are the building blocks of proteins?
Amino acids.

2. Why does fiber pass through us?
It is cellulose, which our bodies cannot digest.

3. Where do nutrients go into the blood?
The small intestine.